COL2010: Data Science

End-Semester Examination — Module 2: Data Cleaning and Preprocessing

Set G

Time: 3 Hours **Total Marks: 100**

Instructions to candidates.

- This paper contains **15** questions. Attempt **all** questions.
- All numerical answers must retain at least four significant figures unless an exact closed form is requested.
- All claims of the form "prove", "show", or "derive" require a fully rigorous argument; a correct final formula with no justification earns no credit.
- State any assumption you invoke that is not already given in the question.
- Calculators are permitted; symbolic computer algebra devices are not.

Q1. For weights $w_1, \ldots, w_d > 0$ and $p \geq 1$, define the weighted Minkowski distance $d_p^w(x, y) = \left(\sum_{j=1}^{d} w_j\, |x_j - y_j|^p\right)^{1/p}$, used by a telecom analytics platform to compare subscriber usage-feature vectors.

(a) Prove that d_p^w is a genuine metric on $\mathbb{R}^d$ for every finite $p \geq 1$. (*Hint:* let $x'_j = w_j^{1/p} x_j$; show $d_p^w(x, y) = d_p(x', y')$, the ordinary unweighted Minkowski distance between the rescaled vectors, and invoke the already-established Minkowski triangle inequality for d_p.)

(b) For $p = 2$, prove precisely that $d_2^w(x, y) = d_M(x, y)$, the Mahalanobis distance with covariance matrix $\Sigma = \mathrm{diag}(w_1^{-1}, \ldots, w_d^{-1})$, i.e. weighted Euclidean distance is exactly diagonal-covariance Mahalanobis distance. State the precise correspondence between the weights w_j and the entries of Σ^{-1}.

(c) Explain briefly why, for d_p^w with $p \neq 2$, there is in general **no** corresponding Mahalanobis-style "covariance" interpretation, and state which property of the $p = 2$ case is essential to that interpretation.

[8 marks]

Q2. Call-duration readings (minutes) for $n = 9$ calls on one subscriber line are:

$$3.2,\ 3.5,\ 3.6,\ 3.8,\ 3.9,\ 4.0,\ 4.2,\ 4.5,\ 45.0.$$

Compute Q_1, the median, Q_3, and IQR via the median-of-halves convention, and hence the Tukey fences. State which observation(s) are flagged. **[4 marks]**

Q3. A telecom billing system suffers a glitch: a fraction $\varepsilon \in (0, 1)$ of dropped calls are logged with an identical placeholder duration M (a default fault code), while the remaining calls have genuine durations confined to a bounded range negligible relative to M as $M \to \infty$.

(a) Derive $\mu(M)$, $\sigma(M)$, and show the ordinary z-score of a placeholder call converges to $\sqrt{(1 - \varepsilon)/\varepsilon}$ as $M \to \infty$.

(b) Solve for the critical fraction ε^* at which this limit equals 3, and state the operational conclusion: what fraction of a billing batch being corrupted identically renders the classical 3σ rule permanently unable to flag it, however extreme M is?

(c) Prove that the modified z-score based on median/MAD instead diverges to $+\infty$ as $M \to \infty$ for any fixed $\varepsilon < 0.5$. State precisely, and distinguish clearly, the two different senses of "breakdown point" at play here, and show they give numerically different conclusions for this billing batch.

[9 marks]

Q4. Two subscribers are described by (Monthly Data Usage in GB, Monthly Spend in paise): $P = (1.8, 38000)$, $Q = (4.5, 71000)$.

(a) Compute the raw Euclidean distance and identify the dominating coordinate.

(b) Given cohort statistics $\mu_{\text{data}} = 3.0, \sigma_{\text{data}} = 0.9$ and $\mu_{\text{spend}} = 52000, \sigma_{\text{spend}} = 10000$, standardize both coordinates and recompute the Euclidean distance.

(c) A churn-risk k-NN model is run on raw features. Explain, quoting your two distances, why recording spend in paise rather than rupees would silently make data usage irrelevant to the churn-risk neighbour search.

[6 marks]

Q5. Two subscribers' logged service-issue tags over {CallDrop, SlowData, Billing, SIMIssue, NetworkOutage} are $x = (1, 1, 0, 0, 1)$ and $y = (1, 0, 0, 1, 1)$.

(a) Compute the Jaccard similarity and cosine similarity of x and y.

(b) A third subscriber's tag vector $z = (0, 0, 0, 0, 0)$ reflects a support ticket that was never categorized, not a genuine claim of zero issues. State $\cos(z, y)$ and $\cos(z, z)$ under the course convention, and explain why silently treating z as genuine "no issues" would corrupt an issue-similarity-based support-routing tool.

[4 marks]

Q6. Let R be the missingness indicator for a column $Y = (Y_{\text{obs}}, Y_{\text{mis}})$, with mechanism $P(R \mid Y_{\text{obs}}, Y_{\text{mis}}; \phi)$.

(a) State Rubin's formal definitions of MCAR, MAR, and MNAR.

(b) Prove the ignorability result under MAR (with distinct parameters), showing the observed-data likelihood factors cleanly.

(c) A telecom operator observes that subscribers who are already planning to churn are disproportionately likely to skip the optional "reason for reduced usage" exit-survey field, i.e. $P(R = 0 \mid \text{TrueChurnRisk})$ increases with the true (unrecorded) churn risk itself. Classify this mechanism formally, and use your proof in (b)–(c) to explain concretely why no imputation using only observed fields can, even in principle, remove the resulting bias in average disclosed reasons.

(d) Suppose the operator additionally has an always-observed count of network-quality complaints per subscriber, correlated with true churn risk. Explain, in general terms, how conditioning on this variable could move the analysis toward a recoverable MAR-type situation relative to the enlarged observed set, and why this shows the classification is relative to the observed variable set, not intrinsic to churn risk itself.

[8 marks]

Q7. An analyst applies iterative predictive imputation to five subscriber records over Data Usage (GB), Call Minutes, and Monthly Spend (Rs.):

Row	DataUsage	CallMinutes	Spend
1	2.0	200	450
2	3.5	150	**?**
3	1.2	300	380
4	**?**	100	600
5	4.0	250	520

Missing cells are initialized with the column mean over observed values. The cyclic order is **Spend** $\rightarrow$ **DataUsage**, using the fitted relations

$$\widehat{\text{Spend}} = 50\cdot\text{DataUsage}+0.8\cdot\text{CallMinutes}+50, \qquad \widehat{\text{DataUsage}} = 0.01\cdot\text{Spend}-0.005\cdot\text{CallMinutes}+1.$$

(a) State the two initial fill values.

(b) Perform one full iteration (Row 2's Spend, then Row 4's DataUsage), showing all substituted numbers.

(c) Explain, referencing the specific predictor columns involved, why a second iteration leaves both values unchanged here.

(d) Your computed DataUsage for Row 4 exceeds every other observed value in the table by a wide margin. Explain what this reveals about the risk of trusting a linear predictive-imputation model outside the joint range of predictor values it was effectively calibrated on, and state one concrete diagnostic check a practitioner should run before accepting such an imputed value.

[6 marks]

Q8. A subscriber-identity-resolution system linking prepaid and postpaid billing platforms computes pairwise similarities for three subscriber records A, B, C: $s(A, B) = 0.79$, $s(B, C) = 0.77$, $s(A, C) = 0.14$, with lower threshold 0.4 and upper threshold 0.70.

(a) Classify each pair.

(b) State the connected-components clustering of $\{A, B, C\}$ under "match" edges, and explain the transitivity problem.

(c) With $\text{Obj}(\mathcal{C}) = \sum_{i,j \text{ same cluster}}(s(i, j)-\theta)$, $\theta = 0.5$, compute Obj for $\{A, B, C\}$ versus $\{A, B\}, \{C\}$, and conclude which is objectively preferable.

(d) For a chain of n subscriber records with adjacent similarity s_0 (just above upper threshold) and non-adjacent similarity floor f (below lower threshold), derive Obj exactly for (i) the single connected-component clustering and (ii) the clustering retaining only adjacent pairs, and prove the gap is $\Theta(n^2)$. State the operational lesson for an operator merging subscriber identities across many years of independently migrated billing platforms.

[10 marks]

Q9. Monthly data-overage charges (in Rs.) for nine subscribers are strongly right-skewed: 4, 7, 11, 18, 28, 50, 95, 240, 980. Compute $\log_{10}$ of each to two decimal places, and state the raw ratio max / min against the transformed range, quantifying the compression achieved. **[4 marks]**

Q10. A telecom analytics team fits a scaler for *Subscription Age (months)* and a one-hot encoder for *Handset Brand* on the entire dataset (training and test rows combined) before splitting, to predict churn.

(a) With $n_{\text{tr}}, n_{\text{te}}$ the train/test sizes and $\mu_{\text{tr}}, \mu_{\text{te}}$ the true population Subscription-Age means restricted to each subpopulation (which may differ if the test period follows a major promotional campaign), derive the exact bias of the leaked full-data mean relative to the correct train-only mean, and state the condition under which it vanishes.

(b) Explain formally why this bias implies any downstream test-set churn-prediction metric is optimistically biased.

(c) A newly launched Handset Brand appears only among test-period subscribers. State precisely how a leak-free encoder fitted on training brands only must represent this row, and explain why fitting the encoder on the union of all brands seen across the whole dataset is still leakage.

[8 marks]

Q11. Two candidate duplicate subscriber records are compared on four fields: name similarity $= 0.93$ (Jaro–Winkler), IMEI-last-4 agreement $= 1$, secondary-phone agreement $=$`NaN` (missing on one record), address agreement $= 0$. Weights: name 0.4, IMEI 0.3, phone 0.2, address 0.1.

(a) Compute the weighted-mean score using only available fields, weights renormalized.

(b) Classify against thresholds lower $= 0.5$, upper $= 0.75$.

(c) Explain why substituting 0 for the missing phone field instead of excluding and renormalizing would bias the decision, and in which direction.

[6 marks]

Q12. Call-setup latency (milliseconds) for $n = 7$ sampled calls: 120, 125, 118, 130, 120, 125, 900.

(a) Compute the mean, standard deviation, and ordinary z-score of the value 900.

(b) Compute the median, MAD, and modified z-score of 900.

(c) Using $|z| \geq 3$ and $|z_{\text{mod}}| > 3.5$, state which method(s) flag 900, and connect the disagreement explicitly to your general result in Q3.

[4 marks]

Q13. For $\mu \in \mathbb{R}^2$ and $\Sigma \succ 0$, define $f(x) = (x - \mu)^\top \Sigma^{-1}(x - \mu)$, so the Mahalanobis "normal usage region" used by a churn-anomaly detector is the sublevel set $E_c = \{x : f(x) \leq c^2\}$.

(a) Prove, directly from the definition of convexity (i.e. show $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y)$ for all x, y and $\lambda \in [0, 1]$, *without* invoking the Hessian), that f is a convex function on $\mathbb{R}^2$. (*Hint:* write $x - \mu = u$, $y - \mu = v$, expand $f(\lambda x + (1 - \lambda)y)$ in terms of u, v, Σ^{-1}, and use positive-definiteness of Σ^{-1} applied to $u - v$.)

(b) Using part (a), prove that E_c is a convex set for every $c > 0$.

(c) Explain, in the language of subscriber usage profiles, why convexity of E_c guarantees that the *average* of any two usage profiles both classified as "normal" is itself classified as "normal" — a stability property that an arbitrary non-convex black-box anomaly detector need not satisfy.

(d) Explain precisely what happens to E_c if Σ is only positive *semi*-definite (singular) rather than positive definite — i.e. some direction has zero variance — and why the "ellipsoid" description then breaks down.

[8 marks]

Q14. Two blocking passes generate candidate duplicate subscriber pairs among $n = 6000$ standardized records linking prepaid and postpaid systems. Pass 1 (same registered circle and birth year) yields blocks of sizes $52, 68, 33$. Pass 2 (same last-four IMEI digits) yields blocks of sizes $10, 13, 8, 17$. The two candidate sets overlap in 55 pairs.

(a) Compute the candidate pairs from each pass and the size of their union.

(b) Compute the reduction ratio RR for the union relative to $\binom{n}{2}$.

(c) Ground truth contains 340 true duplicate pairs; Pass 1 alone captures 290, Pass 2 alone captures 190, both jointly capture 150. Verify via inclusion–exclusion the pair completeness of the union and state its numerical value.

(d) Prove $1-(1-p_1)(1-p_2) \geq \max(p_1, p_2)$ for independent capture probabilities $p_1, p_2 \in [0, 1]$, and relate it to your answer in (c).

[6 marks]

Q15. A telecom data-quality officer gives you eight subscriber records:

ID	Sub. Age (mo.)	Data (GB)	Call Min.	Spend (Rs.)	Name (standardized)
1	34	2.10	220	450	arjun mehta
2	29	2.00	200	420	arjun mehta
3	45	2.30	240	?	neha kapoor
4	51	2.60	260	610	vikram rao
5	38	9.50	30	590	farida sheikh
6	60	2.40	250	?	d. bhatt
7	41	2.35	245	580	neha kapoor
8	33	2.15	225	460	priya sen

Among the six rows with observed Spend, the mean Subscription Age is 37.67; among the two with missing Spend, the mean Subscription Age is 52.5. From the other seven rows (excluding Row 5), $\mu = (\mu_D, \mu_C) = (2.29, 234.3)$ for (Data, CallMinutes) and $\Sigma = \begin{bmatrix} 0.045 & 1.8 \\ 1.8 & 380 \end{bmatrix}$. Rows 3 and 7 are suspected duplicate subscriber records, with comparison vector: name similarity 0.95, IMEI `NaN`, circle agreement 1, DOB agreement 0, using the weights/thresholds of Q11.

(a) State, with the standard caveat about MNAR being unprovable from observed data alone, what the age-conditioned evidence does and does not establish about the missingness mechanism of Spend.

(b) Compute, by hand, the Mahalanobis distance of Row 5's (Data, CallMinutes) = $(9.50, 30)$ from μ, explicitly computing Σ^{-1}, and comment on whether Row 5 is a multivariate outlier despite each coordinate arguably looking merely "unusual" rather than impossible in isolation.

(c) Decide, via the Q11 machinery, whether Rows 3 and 7 should be merged, and construct the merged record's Spend field using the course's numeric golden-record rule.

(d) Argue formally whether deduplication should precede or follow imputation here, prove your ordering strictly dominates the reverse for at least one field, and state the audit-trail invariant that must survive regardless of ordering.

[9 marks]